

Tenure data gameplan (mutually agreed plan)

What this file is: The **forward-looking contract** for what we will accomplish in the **PEER** empirical setting (CS faculty tenure at R1 institutions) and **how** it connects to the inverted-U thesis (Army CODA, basketball SCOUT, academia). It is **strategic**: outcomes, stages, durable artifacts, and advisor alignment. **Algorithmic detail**—how each notebook cell reads and writes specific `.csv` / `.jsonl` / `.py` files—is in `TENURE_PIPELINE_OVERVIEW.md` (the “pipeline map”). `540_tenure_pipeline.ipynb` is the **conductor** at the workspace root; heavy logic lives in `tenure_pipeline/` modules where possible.

Lineage: Narrative builds on `PertinentThoughtsTenure.md`, prior PEER design notes, and the live pipeline described in `TENURE_PIPELINE_OVERVIEW.md`. The sports parallel is `../sports_documents/SPORTS_DATA_GAMEPLAN.md`: same split—**gameplan = intent and stage contract, overview + notebook = implementation wiring**.

Working agreement (process)

- Gameplan vs overview**: `TENURE_DATA_GAMEPLAN.md` (this file) stays **stable and readable**—goals, stage boundaries, durable files, open questions. `TENURE_PIPELINE_OVERVIEW.md` carries **cell-level** behavior, flags, checkpoints, and file schemas as the implementation reference. For **stable handles** to each `##` section here (`§G1–§G14` + front matter), see `TENURE_PIPELINE_OVERVIEW.md §0`.
- Document first for big design changes**: If a stage’s *meaning* changes (e.g. tenure vs attrition definition), update **this gameplan** (or an advisor memo), then align code/notebook.
- Snapshots**: The repo and `tenure_pipeline/` hold truth; obsolete or one-off experiments should live under clearly named archive folders, not mixed into canonical outputs.
- Panel time semantics & collisions (implementation detail)**: Use **exact Wayback timestamps** (derive e.g. `snpsht_dt`) as the **analysis clock** for ordering and `merge_asof`; treat `year` / `season` as **CDX culling strata**, not substitutes for observation time. For **duplicate or ambiguous names**, allow **multiplicity, blending, and conflict flags** for later QA—`faculty_linker` is a **baseline** path, not the only defensible representation. See `TENURE_PIPELINE_OVERVIEW.md §5` (subsections *Temporal keys* and *Collisions*).
- Fix plumbing early**: Structural mistakes in scrape/plan/panel **do not get clearer with age**. Prefer correcting **identity, retention, and join logic** close to when we touch that stage—then re-run—rather than “someday” after results exist.

Snapshot identity, retention, and season labels

Primary key for a Wayback observation is the **Archive capture timestamp** (and the corresponding `wayback_url` / derived `snpsht_dt`). `year` + `season` (spring/fall) are **useful derived labels**—e.g. for heatmaps, stratification, and CDX query bands—but they are **not** a substitute for a continuous timeline.

Retention: Through ingest and parse, **do not discard** captures merely because another capture exists in the same nominal season. Prefer **many dated observations; algorithmic consolidation** (median week, dedupe rules, event logic) is a **downstream** step with logged assumptions—not an early filter.

CDX / plan (Cells 3A–3B): The pipeline requests **up to** `CDX_SNAPS_PER_SEASON` **distinct captures per (calendar year, season band)** (default 12 in CELL 0; see overview). When more CDX rows exist than that cap, the closest `CDX_SPACING_POOL_MULT` × cap to the season anchor are pooled, sorted in time, then **evenly subsampled** so captures are spread across the year. Each selected capture has a **unique on-disk name** including the Wayback `timestamp`. **More snapshots is better** until a later phase explicitly summarizes; overnight runs are acceptable.

What counts as “tried” for CDX: CELL 3A builds `tried_urls` **only from** `faculty_snapshots.plan.jsonl` (plus `tried_primary_url` / `tried_urls` fields on prior rows). Deleting or archiving that file (and, for a full Wayback reset, `cdx_retry_queue.jsonl`, `faculty_snapshots.index.jsonl`, `faculty_snapshots/`, and downstream parse/panel JSONL) makes URLs **untried** again for the next run. Removing HTML alone does not reset the plan.

3A → 3A-RETRY spacing: The notebook includes an optional **bridge cell** after 3A-VIZ: displays a short **hourglass** message and sleeps `CDX_SLEEP_BEFORE_RETRY_SEC` (default 30 minutes) before you run **CELL 3A-RETRY**, so the Archive gets a breather between pass 1 and the slow retry pass.

Option B on disk (naming): `faculty_snapshots/<uni_slug>/<source_id>/<year>_<season>_<timestamp>.html` (legacy two-part `year_season.html` filenames may still appear until rebuilt).

Target architecture: 540 as pipeline conductor

Intent: `540_tenure_pipeline.ipynb` is the **single entry point** you run for the scrape → parse → (future) match → panel chain. Cells are gated by `RUN_CELL*` flags in Cell 0 (see overview).

Role	Responsibility
<code>tenure_pipeline/*.py</code>	Parsers (<code>html_parser.py</code>), URL helpers (<code>apply_url_updates.py</code>), utilities; importable, testable units.
<code>540</code> notebook	Orchestration: load constants, run stages in order, checkpoint/resume, thin cells that call into <code>.py</code> where factored.
Human loops	URL research → <code>url_update_worksheet.csv</code> → <code>apply_url_updates.py</code> → <code>r1_schools_data.py</code> ; optional QA CSVs for match overrides when entity resolution lands.

Why this fits: Matches **document-first** staging, **append-only** plan/index JSONL, and your preference for **one conductor** instead of scattered one-off notebooks.

Outcomes we want

- **Empirical:** A defensible longitudinal panel of CS faculty at R1 departments with **rank / presence** from Wayback snapshots and **publication-based performance** from bibliographic data (OpenAlex and/or DBLP—see stages below).
- **Thesis-facing:** Measures that support an **inverted-U** test: tenure or promotion success vs **leave–self-out peer pool quality** (and related peer-group definitions), analogous to Army and basketball designs.
- **Operational:** Reproducible **HTML archive** per planned snapshot, **parsed** roster records, then **matched** people to author IDs, then **panel** rows with explicit **sample-loss accounting** at each filter (advisor: “track how many people you throw away at every point”).
- **Coverage:** Full R1 CS roster in the plan is a **publication-time** goal; **near-term** priority (advisor, Apr 2026) is **quality and end-to-end statistics on the current corpus**, not maximizing scrape breadth every week.

Durable team / URL equity (files on disk)

Intent: Treat these as **durable inputs**, not disposable cache—extend when QA shows gaps.

File (location)	Role
tenure_pipeline/r1_schools_data.py	Source of truth for PILOT_SCHOOLS and per-school urls[] .
tenure_pipeline/r1_cs_departments.csv	CSV emitted by Cell 2 from the Python list—downstream convenience.
tenure_pipeline/url_update_worksheet.csv	Human-edited queue for new/changed URLs; drives apply_url_updates.py .
tenure_pipeline/faculty_snapshots_plan.jsonl	Append-only CDX plan + sentinels (n_snaps , bookmarks, reasons)—see overview for semantics; sole driver of “tried” URLs for 3A skip logic.
tenure_pipeline/cdx_retry_queue.jsonl	URLs that timed out in 3A pass 1; consumed by CELL 3A-RETRY (clear when doing a full clean slate).
tenure_pipeline/faculty_snapshots_index.jsonl	Per-download audit (HTTP status, bytes, paths).
tenure_pipeline/faculty_snapshots_parsed.jsonl	Parsed faculty rows (Cell 4).
tenure_pipeline/faculty_panel.jsonl	Longitudinal within-school panel + Wayback timestamps (Cell 5); see overview §5.
tenure_pipeline/faculty_panel_collisions.jsonl	Name-key collision QA (Cell 5).
tenure_pipeline/dblp_parsed/*.jsonl	Per-year DBLP extracts (Cell 1).
tenure_pipeline/school_enrollment_annual.csv	IPEDS fall headcount by school × year (for viz; rebuild 541 / build_school_enrollment_from_ipeds.py).
tenure_pipeline/openalex/*.json / *.jsonl	Cells 6A–6B outputs when run (institution map, author IDs, works-by-year, low-confidence queue).
tenure_pipeline/faculty_panel_enriched.jsonl	Cell 7: pubs + tenure/attrition/censoring flags.
tenure_pipeline/faculty_panel_with_pools.jsonl	Cell 8: LOO peer pool metrics (poolq_loo_mean , etc.).
tenure_pipeline/stage9_inverted_u.png / stage9_binned_table.csv	Cell 9: preliminary inverted-U artifacts.
tenure_pipeline/faculty_panel_advisor.csv / R1_tenure_data.csv	Cell 543: flat CSV for advisors / committee (PANEL_CSV_GLOSSARY.md).

Option B on disk: faculty_snapshots/<uni_slug>/<source_id>/<year>_<season>_<timestamp>.html (multi-capture) or legacy <year>_<season>.html ; source_id = faculty_source_id(url) —see overview.

Stage map (high level)

Stage	Notebook region	Intent	Key outputs	Detail
0	Cell 0	Flags, paths, imports	—	Overview §2
1	Cell 1	DBLP XML → yearly JSONL	<code>tenure_pipeline/dblp_parsed/</code>	Overview §3
2	Cell 2	Schools list → CSV	<code>r1_cs_departments.csv</code>	Overview §3
3A–3E	Cells 3A–3E (+ optional bridge before 3A-RETRY)	CDX plan, Wayback download, optional rescues	<code>faculty_snapshots_plan.jsonl</code> , HTML tree, <code>cdx_retry_queue.jsonl</code> , indexes	Overview §3–4
4	Cell 4	Parse HTML → names/ranks	<code>faculty_snapshots_parsed.jsonl</code>	Overview §4
5	Cell 5	Within-school longitudinal panel + Wayback plan join (<code>snpsht_dt</code> , etc.)	<code>faculty_panel.jsonl</code> , <code>faculty_panel_collisions.jsonl</code>	Overview §4
6A–6B	Cells 6A–6B	OpenAlex institution + author resolution + works-by-year (API)	<code>openalex_*.json</code> / <code>*.jsonl</code>	Overview §4
7	Cell 7	Enriched panel (pubs + tenure/attrition events)	<code>faculty_panel_enriched.jsonl</code>	Overview §2
8	Cell 8	LOO peer pool metrics	<code>faculty_panel_with_pools.jsonl</code>	Overview §2
9	Cell 9	Binned inverted-U check	<code>stage9_inverted_u.png</code> , <code>stage9_binned_table.csv</code>	Overview §2; §G12
10+	Cells 10, 10.5, 12 (+ 543 )	Layer B Cox survival (planned) + advisor CSV package (done)	<code>faculty_panel_advisor.csv</code> ; Cox artifacts TBD	Overview §4

Full sentinel rules, CDX timeouts, 3B ghost retries, and parser strategies are **not** duplicated here—see `TENURE_PIPELINE_OVERVIEW.md` .

Advisor direction — 2026-04-09 conversation (Alex Gates)

- Source: `2-Way_Ahead/20260410_Paper_directions_4_otter_ai.pdf` (Otter transcript). Summarized for implementation alignment.
- 1. **Priority shift (next ~week):** The scrape/plan machinery is in good shape. **Do not** treat “add more schools / URLs” as the main task **right now**. Shift time toward **quality and measurement on existing data**: how many **assistant** → **associate** transitions you can see, how much **attrition** (assistant disappears without associate), and rough **peer-group** definitions you can actually populate.
 - 2. **You do not need the full 187-school scrape today**—you need enough structure to estimate effects and see whether the **inverted-U** is plausible; **defend the sample** at publication time.
 - 3. **Core ingredients to plan explicitly:** (a) **success metric**—tenure / promotion to associate; (b) **peer group(s)**—nested definitions (immediate peers vs broader assistant cohorts—flexible, characterize alternatives); (c) **performance metric**—publications/citations via **OpenAlex** (and ORCID for disambiguation); expect to **try several** specifications.
 - 4. **Faculty-per-snapshot distribution:** Treat as **department pool size**; very large counts may include non-tenure-track listings—**filter later**, but start measuring.
 - 5. **Name matching (v0):** Basic **within-school / within-department** consistency before exotic global merges.
 - 6. **OpenAlex at scale:** Prefer **bulk** OpenAlex (UVA **Connected Data Hub** / HPC: `.csv.gz` table shards) over hand-fixing author IDs for subsets; **do not** manually merge three OpenAlex IDs unless the same rule could apply **everywhere** (otherwise bias). **Status (June 2026):** `build_openalex_cache.py` + `openalex_snapshot_cache.jsonl` implement bulk works-by-year on Rivanna when CDH mount is available; Mac uses `rsync'd` cache (see `HPC_SETUP_CHECKLIST.md` , overview §4). Cell 6A author resolution may still use API when not on HPC.
 - 7. **Sanity check literature:** Use **Dakota**-style tenure/productivity work (and any shared **tenure rates**) only to compare **orders of magnitude**—not as the main theoretical anchor.
 - 8. **Field:** **CS** aligns with DBLP; if linkage rates collapse, **other hard sciences** are acceptable—decide on evidence.
 - 9. **Working style:** Drive toward a **first end-to-end curve** (even with messy data); **log sample loss** at each filter; **re-run the pipeline** after a day of cleaning rather than cleaning for a week without integrated results. (Rough time split discussed: **most** time on getting the model through; **small** slice on new URLs / edge HTML.)

Stage 1 — DBLP parse (Cell 1)

We will keep per-year JSONL under `tenure_pipeline/dblp_parsed/` as the canonical bibliographic spine for author names/years to match against later stages. Analysis window for publications may be a subset of years on disk.

Stage 2 — R1 school list (Cell 2)

We will maintain `r1_schools_data.py` as the authoritative list of departments and `urls[]`, rebuild `r1_cs_departments.csv` from Cell 2 after edits, and use `apply_url_updates.py` + the worksheet for controlled URL changes.

Stage 3 — Wayback scrape (Cells 3A–3E)

We will discover snapshots via CDX (3A), download HTML (3B), and use **optional** rescue cells (3C–3E) where archived pages are shells—documented in the overview. **Goal:** durable HTML + plan/index lines suitable for Cell 4.

Stage 4 — HTML parse (Cell 4)

We will parse all `*.html` under each school directory (Option B + `legacy/`) into `faculty_snapshots_parsed.jsonl`, with strategy audit. **Capture broad titles**; narrow to tenure-track populations at analysis time.

Stages 6–9 — Match, enriched panel, pools, analysis (implemented May–June 2026)

Delivered (see `PEER_Status_Update_for_VECTOR_2026-06-03.md` and overview §2):

Piece	Intent
Entity resolution	Faculty names ↔ OpenAlex (Cells 6A–6B in <code>540</code> ; <code>openalex_resolver.py</code>) with scalable rules; DBLP JSONL for cross-checks; optional QA CSV for overrides. Bulk OpenAlex at UVA when access exists.
Panel	<code>(person × year)</code> (or equivalent) with rank transitions, attrition flags, and publication metrics.
Peer quality	Leave–self–out (or pre-specified) pool metrics on agreed performance measure.
Inference / EDA (Layer A)	Binned tenure rates vs pool quality — stage 9 complete (non-monotone pattern, 18 bins).

Cell numbers and artifacts are documented in `TENURE_PIPELINE_OVERVIEW.md` §2–§4.

Layer B — Cox survival (Cells 10 / 10.5 / 12) — **Planned**

Intent: Time-to-tenure Cox with pool-quality linear + quadratic terms — adapt Army `520` workflow to tenure covariates (`poolq_loo_mean`, `pubs_year`, etc.).

Status (corrected 2026-06-11): `540_tenure_pipeline.ipynb` ends at Cell 9. Layer B is **not wired** yet. **543** advisor CSV export is complete separately.

Piece	Status
Cox / survival (Layer B)	 Planned — build Cells 10 / 10.5 / 12 in <code>540</code> ; template: <code>talent/520_pipeline_cox_working.ipynb</code>
Formal HR output	Not yet — blocked on Layer B build

Fast path — first inverted-U checkpoint (dirty OK)

Purpose: See whether **tenure (or promotion) rate** vs **peer pool quality** shows a **nonlinear** pattern early, before perfecting every linkage. **Rule:** Do not delete canonical under `tenure_pipeline/` while experimenting; write scratch outputs with dated suffixes if needed.

Checklist (adapted from sports gameplan logic) —  ALL COMPLETE as of May 2026

- ✓ **Inputs:** `faculty_panel_with_pools.jsonl` (72 MB) — panel with rank, year, `pubs_year`, `pubs_cumulative`, outcome flags.
- ✓ **Define v0 outcomes:** `tenure_event` / `attrition` / `censored` flags; `gap_tolerance=2` years; implemented in `panel_builder.py`.
- ✓ **Define v0 peer pool:** `poolq_loo_mean` — LOO mean pubs among OpenAlex-matched assistants in same dept-year; implemented in `pool_metrics.py`.
- ✓ **Bins:** 18 equal-width bins of `poolq_loo_mean`; tenure rates with Wilson 95% CI per bin.
- ✓ **Save artifacts:** `stage9_inverted_u.png` + `stage9_binned_table.csv` (18 bins × 17 columns).
- ✓ **Optional — LPM:** Not yet estimated (Layer B Cox is the **planned** main spec — not yet in `540`).

Result: Non-monotone pattern consistent with inverted-U. Peak tenure rates at bins 16–17 (LOO median ~8–9 pubs/yr), with a drop at the very top bin 18 (LOO median ~12.7 pubs/yr). See `PEER_Status_Update_for_VECTOR_2026-06-03.md` for full binned results and caveats.

Open points — Updated June 2026

Resolved since April 2026: - ☒ Panel grain and tenure-clock window: `(faculty_id × year)`, `gap_tolerance=2` years; implemented in `panel_builder.py`. - ☒ Peer-group v0: LOO mean pubs among OA-matched assistants in same dept-year; `pool_metrics.py`. - ☒ OpenAlex as primary pub source (DBLP retained as cross-check spine only). - ☒ First inverted-U check (stage 9 / Layer A): signal present; binned artifacts saved. - ☒ Advisor-facing CSV + glossary produced (`faculty_panel_advisor.csv`, `PANEL_CSV_GLOSSARY.md`). - ☒ External briefing prepared: `advancement_under_constrained_distinction_dakota_feedback_v03.rtf` (June 2026, Dakota / academic-careers committee member).

Still open: - Exact **peer-group robustness** set — alternative definitions (broader cohort, field-weighted, citation-based). - **LPM / logit** sign check (`tenure_event ~ poolq_loo + poolq_loo2 + covariates`). - **Layer B — Cox survival:** build Cells 10 / 10.5 / 12 in `540` (adapt `520` pattern); report HR estimates and inverted-U test on quadratic pool-quality term. - **Attrition vs lateral move:** snapshots cannot distinguish “left academia” from “moved institution” — competing-risk framing hedges this; note as limitation. - **Coverage expansion:** current ~60 usable schools; 168 in `PILOT_SCHOOLS`; publication-time goal is defensible breadth. Prioritize analysis lock-in over URL chasing for now. - **OA bulk access upgrade:** UVA CDH bulk snapshot for author resolution (would improve HIGH confidence rate beyond current ~32%). - **Heterogeneity:** subfield (ML vs systems vs theory), cohort-year, and dept prestige tier effects. - **Prestige covariate:** merge NRC rankings or USNews CS tier into panel for controls.

Document history

- **2026-06-11: Layer B correction** — Cox (Cells 10 / 10.5 / 12) reframed as **planned** (`540` ends at Cell 9); split § “Stages 6–9” from Layer B; removed “Cox wired” from resolved open points.
- **2026-06-08:** Stage map rows **7–10+** updated (Cells 7–9 complete). §G11 heading reframed from “planned” to implemented for Cells 6–9. Open points + fast-path checklist marked complete.
- **2026-06-08:** `PEER_report_to_COMPASS.md` created in `3-Master_Plan/`. `TENURE_STREAMLINING_AND_RESEARCH_PRIORITIES.md` Part 3 rewritten with current panel stats + stage 9 status. `PertinentThoughtsTenure.md` § Stage 9 inverted-U added. Durable files table extended (enriched panel, pools, stage 9, advisor CSV). `PEER_Status_Update_for_VECTOR_2026-06-03.md` coverage corrected (168 schools vs quality filter). fast-path checklist marked complete; resolved items noted; still-open items updated. Stage 9 inverted-U result documented. Cox model (Cells 10/10.5), 543 package notebook, `faculty_panel_advisor.csv`, `PANEL_CSV_GLOSSARY.md`, `PEER_Status_Update_for_VECTOR_2026-06-03.md`, and Dakota briefing (`advancement_under_constrained_distinction_dakota_feedback_v03.rtf`) added to resolved items.
- **2026-04-03 — 2026-04-08:** Earlier iterations of `TENURE_DATA_GAMEPLAN.md` (roster size, sentinels, URL workflow, Cell 3A-RETRY)—superseded in structure by this file; technical detail preserved in `TENURE_PIPELINE_OVERVIEW.md`.
- **2026-04-10: Restructured to mirror `SPORTS_DATA_GAMEPLAN.md`**: split **gameplan** vs **overview**; added **Advisor direction (2026-04-09)** from Otter transcript; condensed duplicate cell-by-cell specs in favor of cross-links to `TENURE_PIPELINE_OVERVIEW.md`.
- **2026-04-10 (later):** Working agreement item 1 points to `TENURE_PIPELINE_OVERVIEW.md` §0 for §G1–§G14 stable anchors to each `##` heading here.
- **2026-04 (later):** Working agreement item 4 summarizes §5 in the overview (*Temporal keys, Collisions*) so the gameplan stays the strategic hook without duplicating that text.
- **2026-04 (later):** Stage map row 5 = Cell 5 panel outputs; durable-files table adds `faculty_panel.jsonl` / `faculty_panel_collisions.jsonl`.
- **2026-04-10: Snapshot identity & retention + fix plumbing early** (working agreement 5); Option B filename pattern allows **timestamp** suffix; CDX multi-capture per season band—see overview §3 / §5.
- **2026-04-11:** CDX cap 12 per band + **temporal spacing** when many candidates; `tried_urls` / clean-slate semantics; `cdx_retry_queue.jsonl` in durable files; **3A→3A-RETRY bridge** (`CDX_SLEEP_BEFORE_RETRY_SEC`, hourglass cell)—see overview §2–§3.
- **2026-04-11 (later):** `PertinentThoughtsTenure.md` shortened: dissertation-facing notes only; stale operational counts and duplicate pipeline detail removed; **NC State** status in overview de-“pending” in favour of re-run guidance.
- **2026-04-12:** OpenAlex **bulk snapshot at UVA** — access **pending**; Stage **6A–6B** documented as **API** path; durable-files table adds `school_enrollment_annual.csv` + `openalex_*`; advisor bullet 6 links to overview §4 + Otter transcript; stage map rows **6A–6B / 7+**; **Stages 6–9** table names OpenAlex + bulk.